AWS Task 4

Task Description:

Launch an EC2 instance (Linux and Windows) along with a web server.

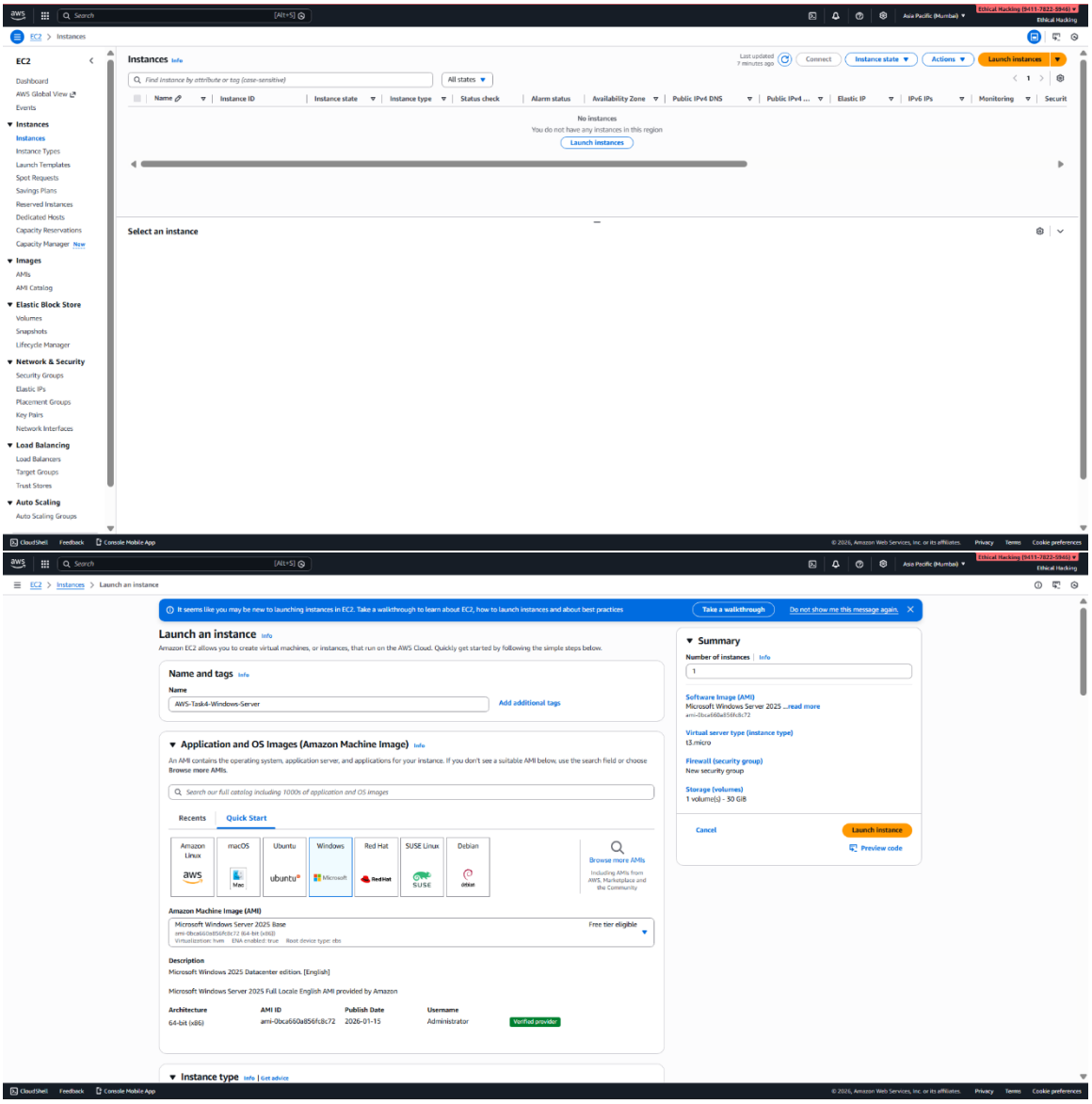
Then, create an EBS volume of 5 GB,

attach it to an EC2 machine (Linux and Windows), and take a snapshot.

Finally, create an EBS volume using the taken snapshot.

Creating Linux and Windows EC2 machines:

Creating Windows Instance:



EC2 > Instances > Launch an instance

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

WindowsEC2 | Create new key pair

For Windows instances, you use a key pair to decrypt the administrator password. You then use the decrypted password to connect to your instance.

Network settings Info

Edit

Network | Info
vpc-00c8ea7cf8e8265ec

Subnet | Info
No preference (Default subnet in any availability zone)

Auto-assign public IP | Info
Enable

Firewall (security groups) | Info
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

- Create security group
- Select existing security group

We'll create a new security group called **'launch-wizard-1'** with the following rules:

- [x] Allow RDP traffic from — Anywhere 0.0.0.0/0
  Helps you connect to your instance
- [ ] Allow HTTPS traffic from the internet
  To set up an endpoint, for example when creating a web server
- [x] Allow HTTP traffic from the internet
  To set up an endpoint, for example when creating a web server

⚠ Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Configure storage Info

Advanced

1x 30 GiB gp3 Root volume, 3000 IOPS, Not encrypted

Add new volume

The selected AMI contains instance store volumes, however the instance does not allow any instance store volumes. None of the instance store volumes from the AMI will be accessible from the instance

Summary

Number of instances | Info: 1

Software Image (AMI)
Microsoft Windows Server 2025 ...read more
ami-0bca660a856fc8c72

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 30 GiB

Cancel | Launch instance | Preview code

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EC2 > Instances > Launch an instance

Default active vCPUs: 2
Total vCPUs: 2

Metadata accessible | Info: Enabled

Metadata IPv6 endpoint | Info: Select

Metadata version | Info: V2 only (token required)

⚠ For V2 requests, you must include a session token in all instance metadata requests. Applications or agents that use V1 for instance metadata access will break.

Metadata response hop limit | Info: 2

Allow tags in metadata | Info: Select

User data - optional | Info
Upload a file with your user data or enter it in the field.

Choose file

```
<powershell>
Install-WindowsFeature -Name Web-Server -IncludeManagementTools

$html = @"
<!DOCTYPE html>
<html>
<head>
<title>Windows EC2 Web Server</title>
<style>
body {
  margin: 0;
  font-family: Segoe UI, sans-serif;
  background: linear-gradient(135deg, #141e30, #243b55);
  color: white;
  text-align: center;
```

- [ ] User data has already been base64 encoded

Summary
Number of instances | Info: 1
Software Image (AMI)
Microsoft Windows Server 2025 ...read more
ami-0bca660a856fc8c72
Virtual server type (instance type)
t3.micro
Firewall (security group)
New security group
Storage (volumes)
1 volume(s) - 30 GiB
Cancel | Launch instance | Preview code

EC2 > Instances

EC2
- Dashboard
- AWS Global View
- Events
- **Instances**: Instances, Instance Types, Launch Templates, Spot Requests, Savings Plans, Reserved Instances, Dedicated Hosts, Capacity Reservations, Capacity Manager New
- **Images**: AMIs, AMI Catalog
- **Elastic Block Store**: Volumes, Snapshots, Lifecycle Manager
- **Network & Security**: Security Groups, Elastic IPs, Placement Groups, Key Pairs, Network Interfaces
- **Load Balancing**: Load Balancers, Target Groups, Trust Stores
- **Auto Scaling**: Auto Scaling Groups

Instances (1/1) Info

Last updated less than a minute ago | Connect | Instance state | Actions | Launch instances

Find instance by attribute or tag (case-sensitive) | All states

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IPv6 IPs	Monitoring
AWS-Task4-Windows-Server	i-06a2a32e170be8989	Running	t3.micro	Initializing	View alarms +	ap-south-1a	ec2-43-205-126-210.ap...	43.205.126.210	–	–	disabled

i-06a2a32e170be8989 (AWS-Task4-Windows-Server)

Details | Status and alarms | Monitoring | Security | Networking | Storage | Tags

Instance summary Info

Instance ID
i-06a2a32e170be8989

IPv6 address
–

Hostname type
IP name: ip-172-31-36-195.ap-south-1.compute.internal

Answer private resource DNS name
IPv4 (A)

Auto-assigned IP address
–

IAM role
–

IMDSv2
Required

Operator
–

Public IPv4 address
43.205.126.210 | open address

Instance state
Running

Private IP DNS name (IPv4 only)
ip-172-31-36-195.ap-south-1.compute.internal

Instance type
t3.micro

VPC ID
vpc-00c8ea7cf8e8265ec

Subnet ID
subnet-0e306588c02279fa9

Instance ARN
arn:aws:ec2:ap-south-1:941178225946:instance/i-06a2a32e170be8989

Private IPv4 addresses
172.31.36.195

Public DNS
ec2-43-205-126-210.ap-south-1.compute.amazonaws.com | open address

Elastic IP addresses
–

AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations. | Learn more

Auto Scaling Group name
–

Managed
false

Instance details Info

AMI ID
ami-0bca660a856fc8c72

Monitoring
disabled

Platform details
Windows

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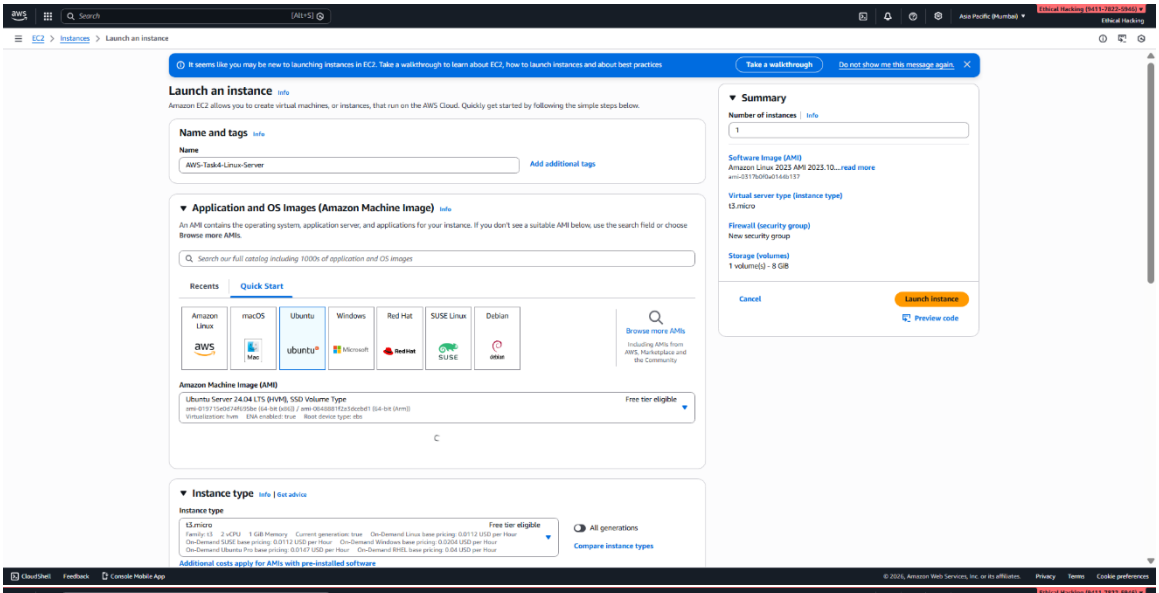
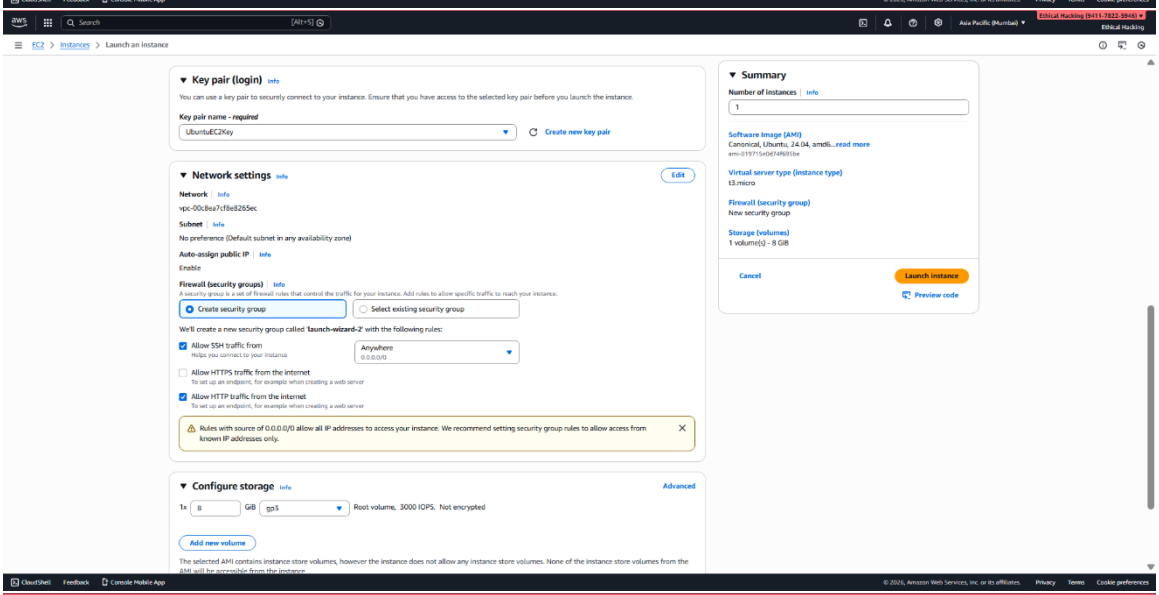
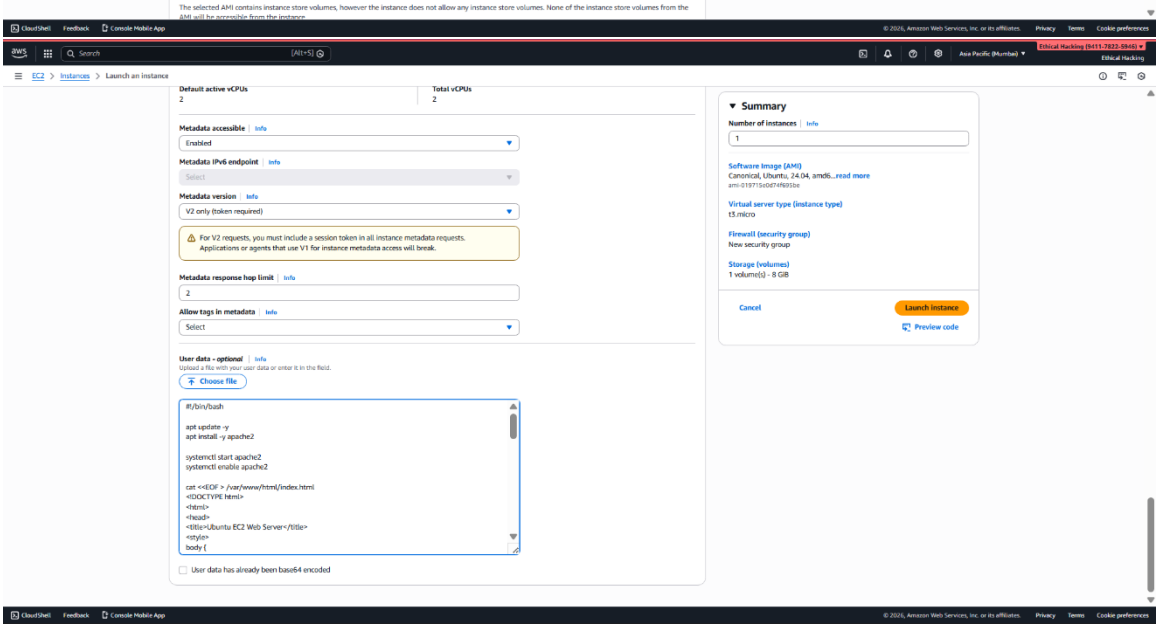
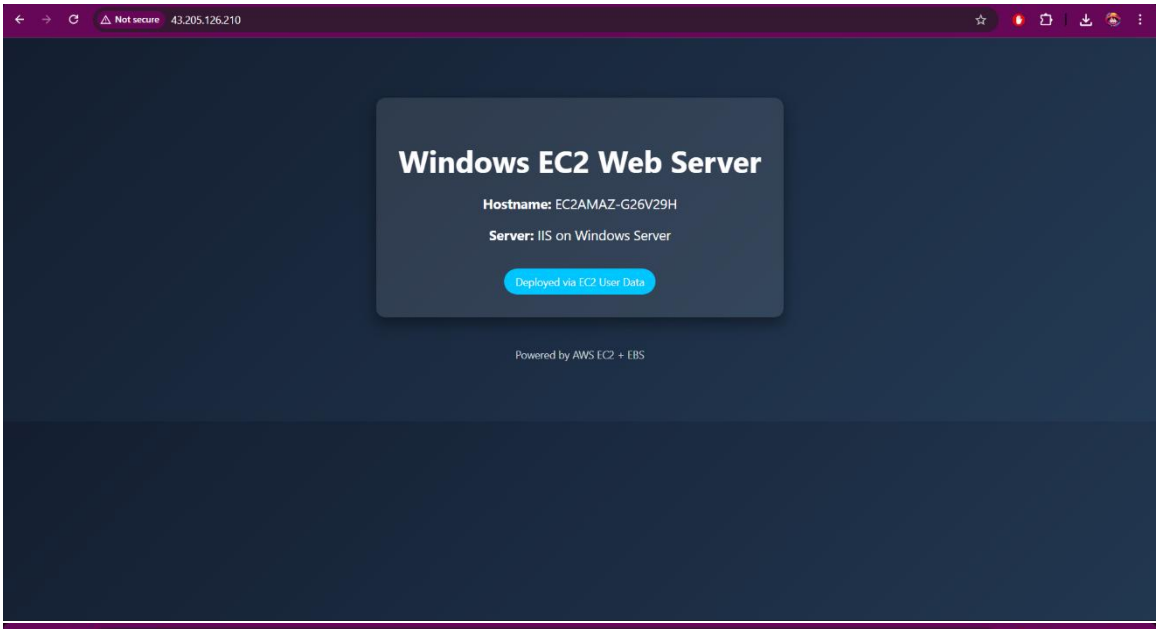
Creating Linux Instance:

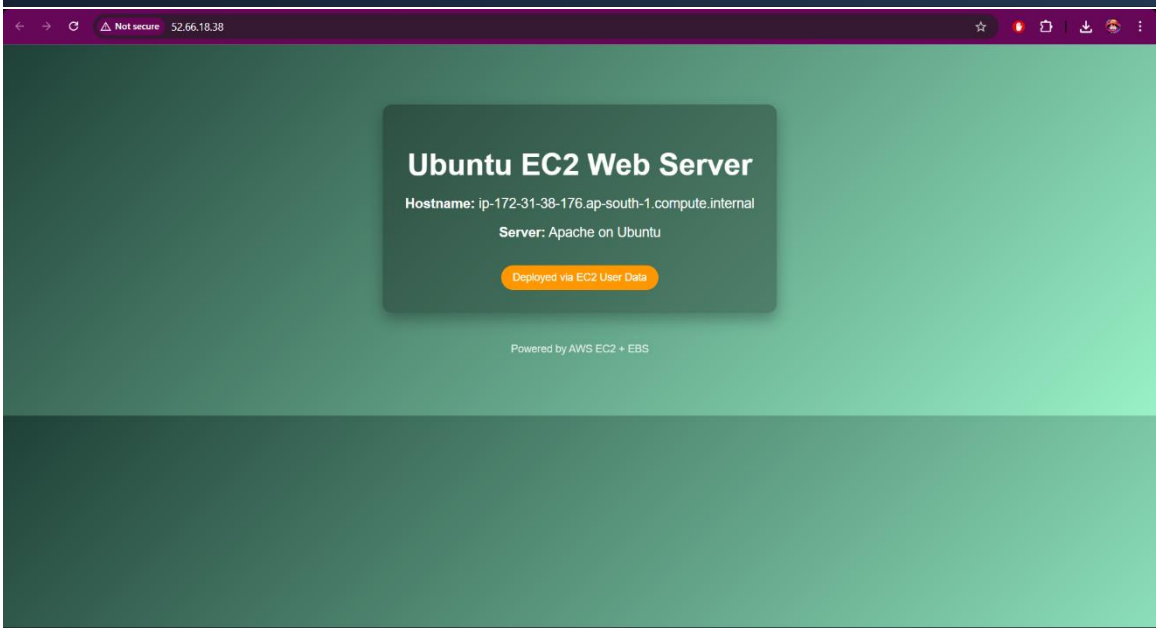






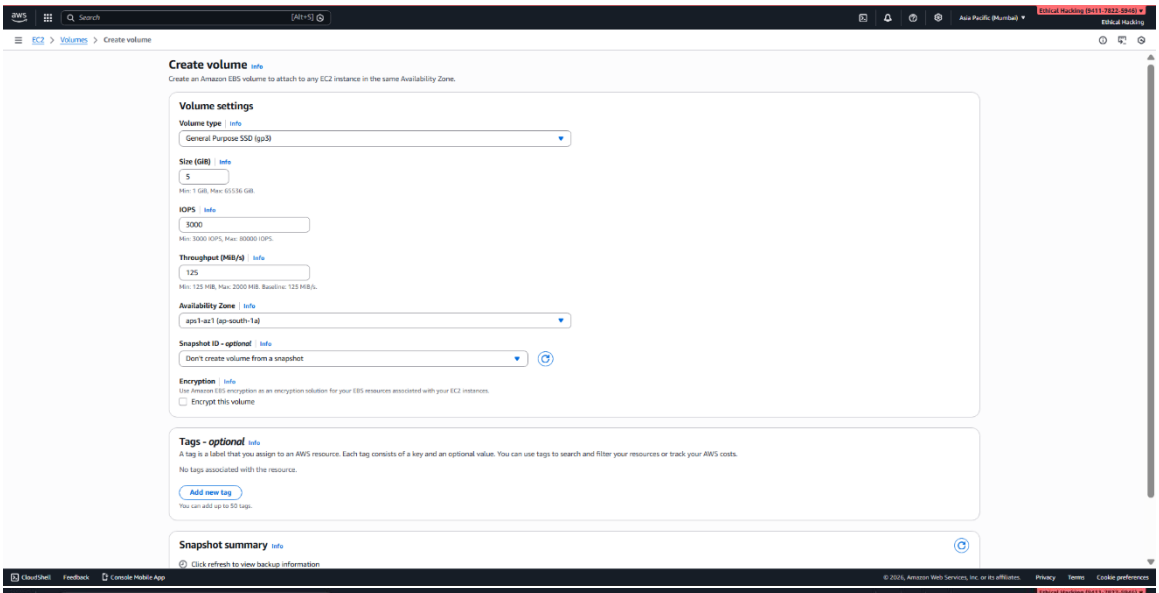
After creating both EC2 machines, validating the server by checking the IP

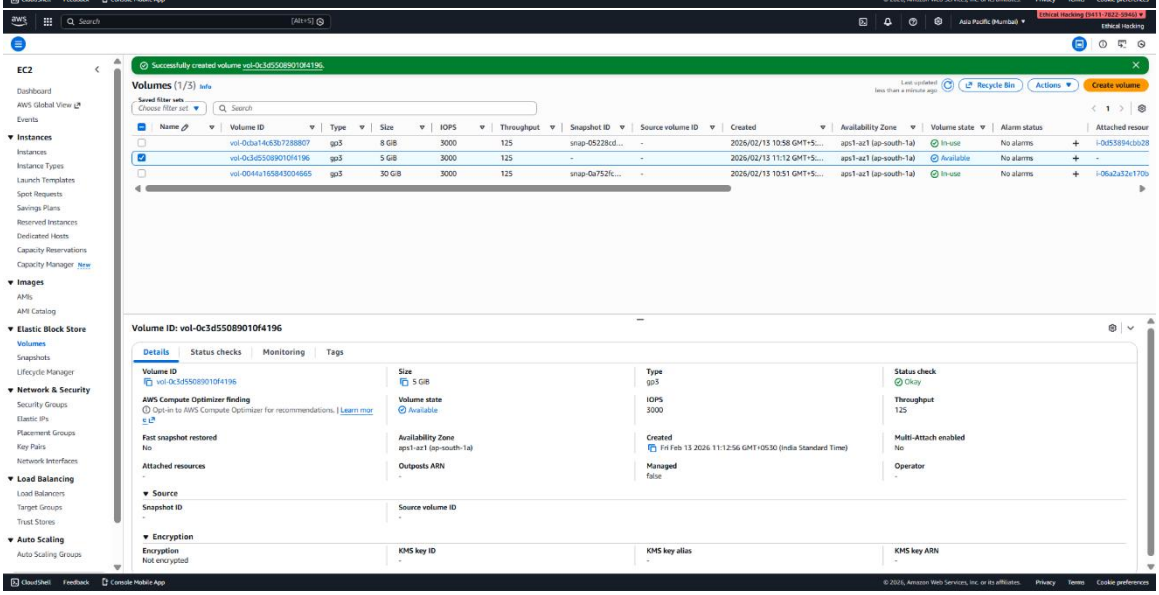




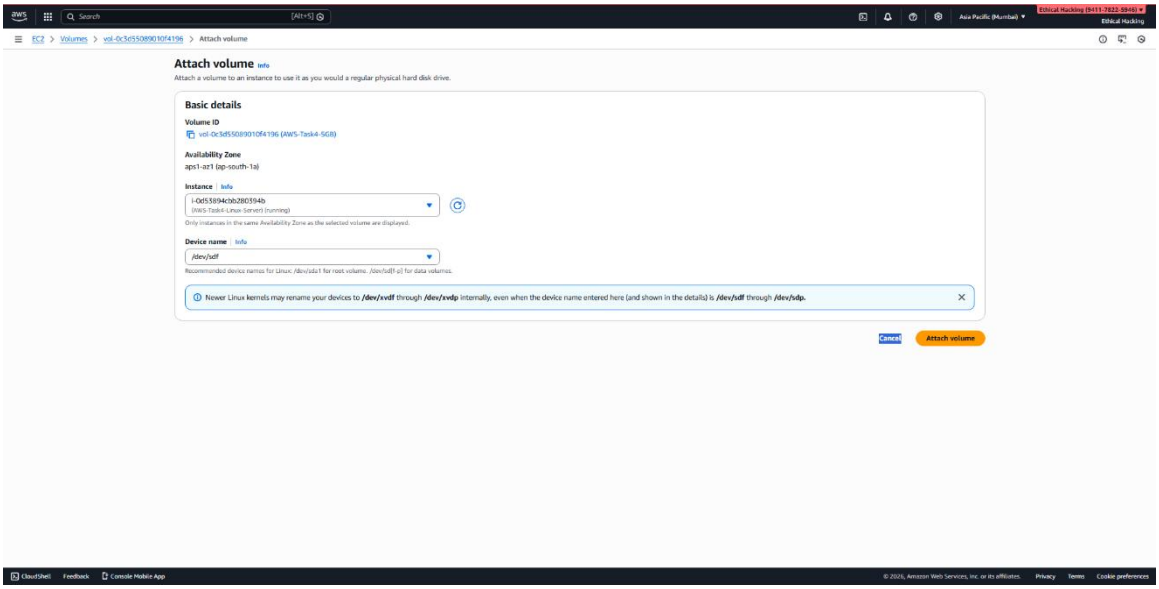
Creating EBS:

Creating volume from Elastic Block Storage → Volumes → Create Volume





After creating the volume, attaching it to the instance:

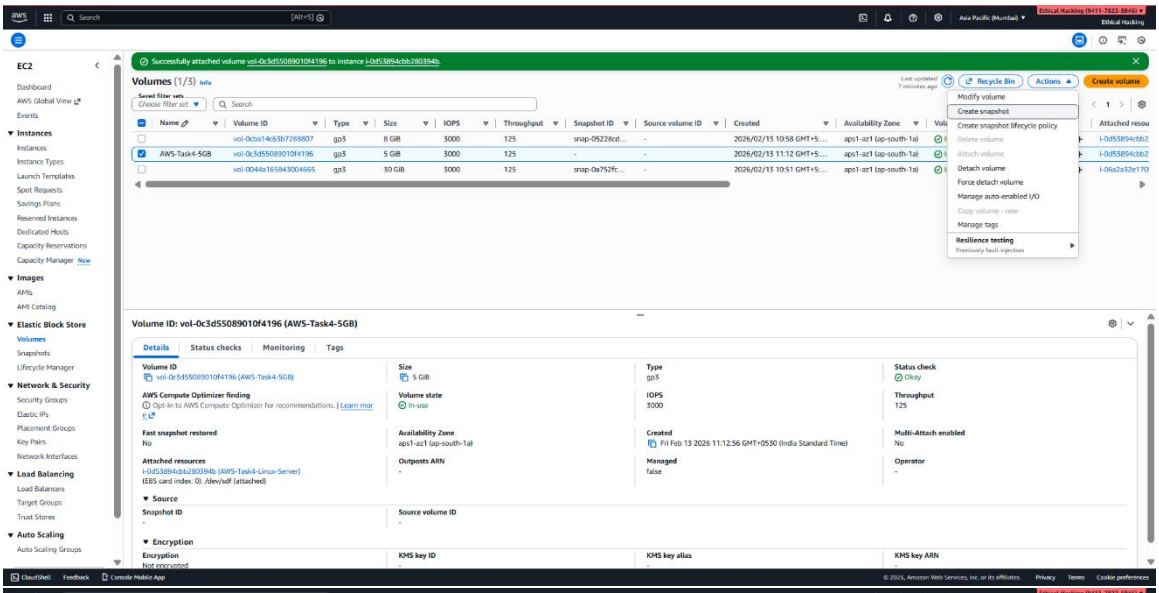


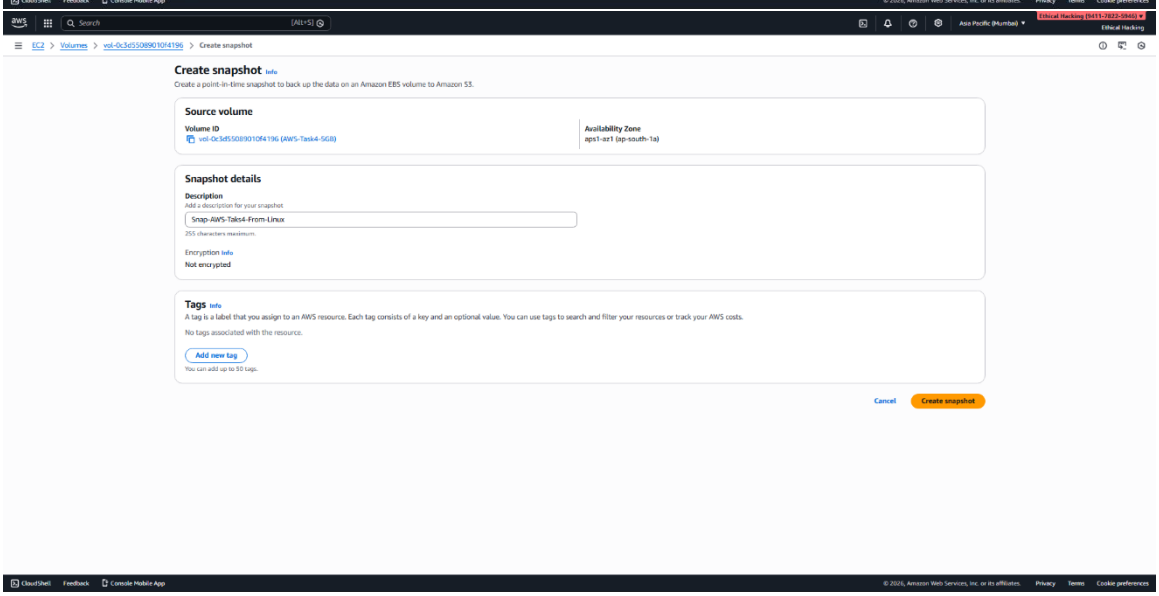
After attaching the volume to the Linux instance, validating if it's been mounted:

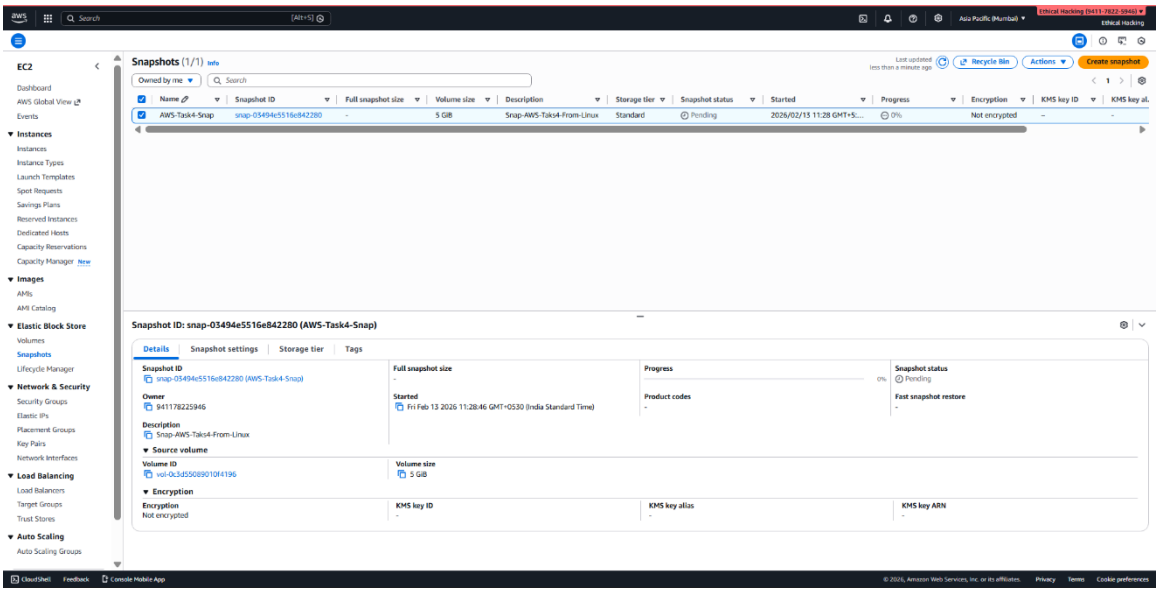
```
ubuntu@ip-172-31-38-176: /d

ubuntu@ip-172-31-38-176:/data/test$ lsblk
NAME         MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0          7:0    0 27.6M  1 loop /snap/amazon-ssm-agent/11797
loop1          7:1    0   74M  1 loop /snap/core22/2163
loop2          7:2    0 50.9M  1 loop /snap/snapd/25577
nvme0n1      259:0    0    8G  0 disk
├─nvme0n1p1  259:1    0    7G  0 part /
├─nvme0n1p14 259:2    0    4M  0 part
├─nvme0n1p15 259:3    0  106M  0 part /boot/efi
└─nvme0n1p16 259:4    0  913M  0 part /boot
nvme1n1      259:5    0    5G  0 disk /data
ubuntu@ip-172-31-38-176:/data/test$
```

Now creating snapshot for this volume:

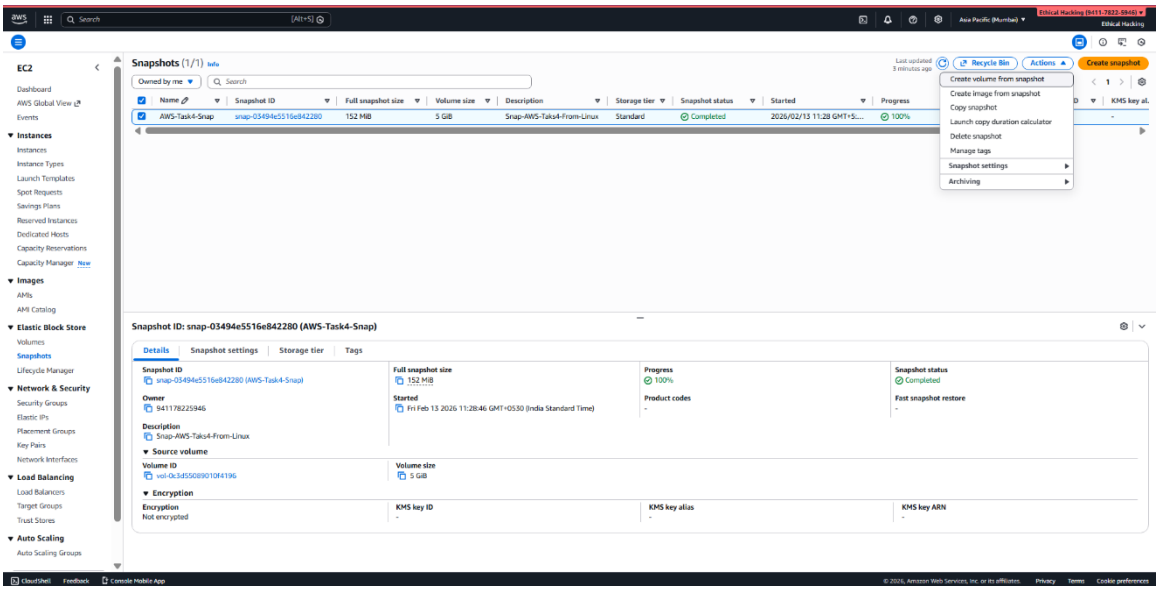




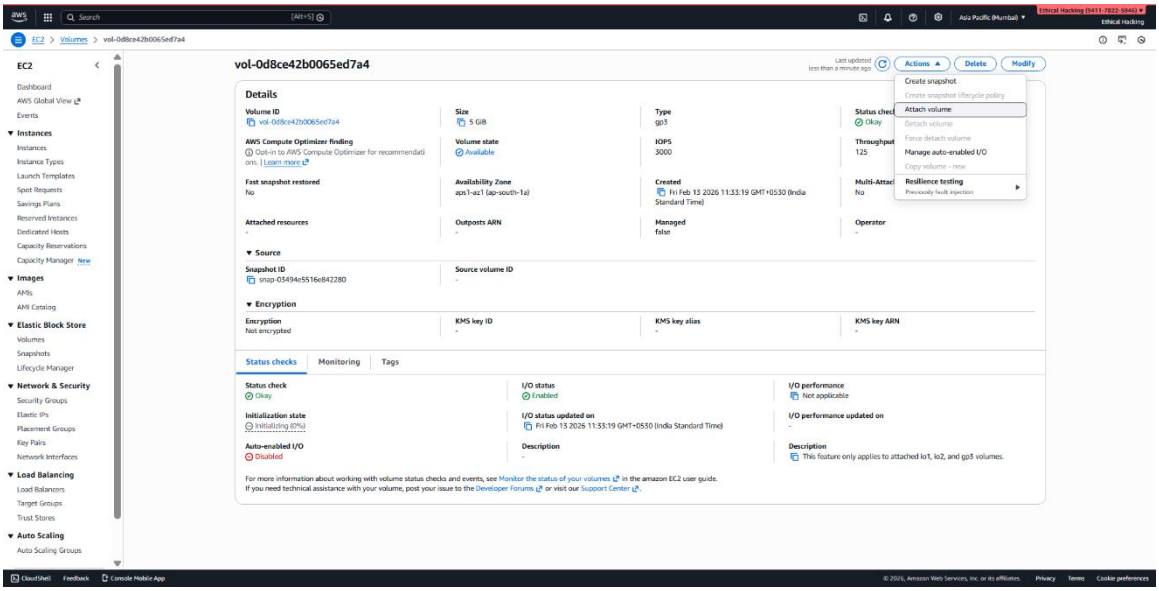


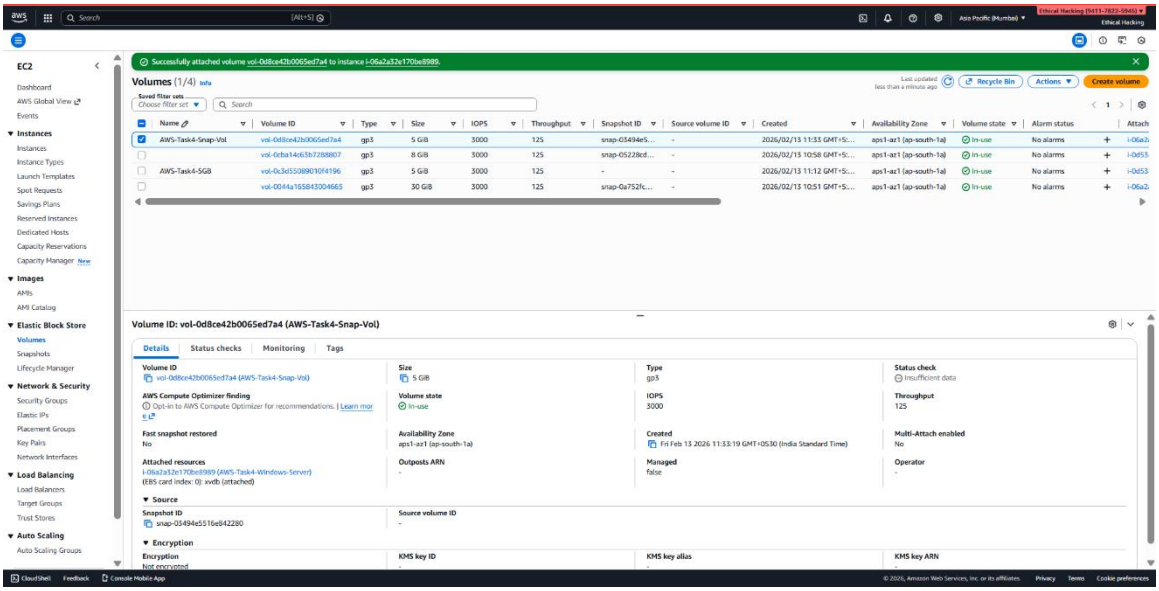
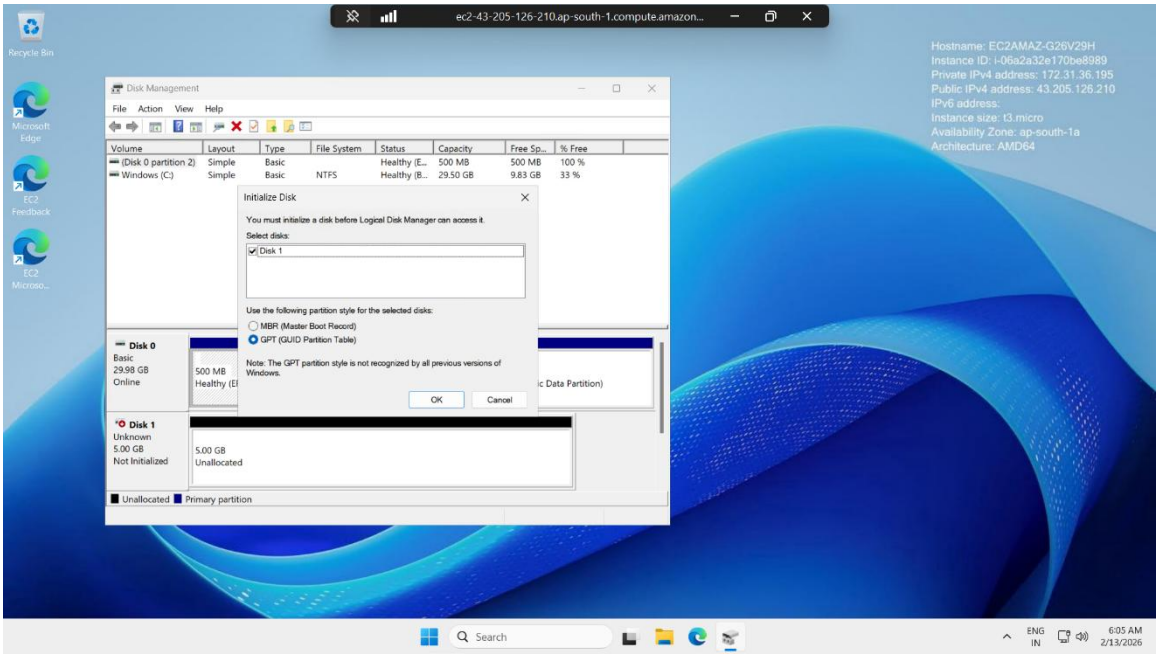
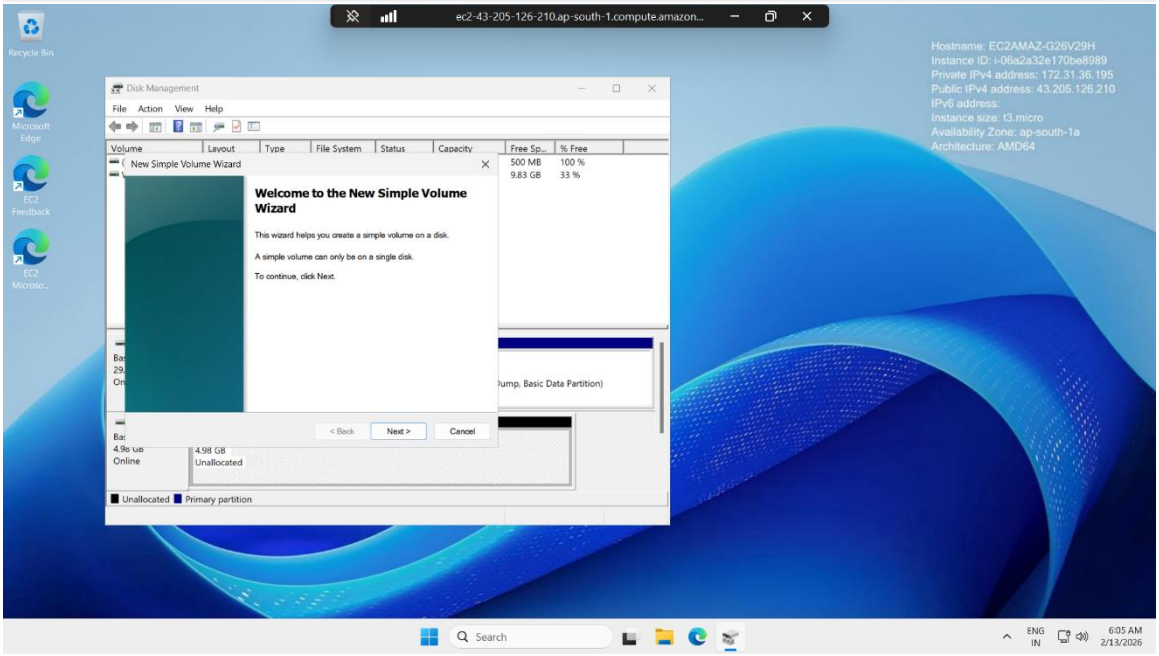
Creating EBS volume from the taken Snapshot:

After creating the snapshot, I am going to create the new EBS volume from this snapshot:



After creating the EBS volume, attaching it to the EC2 machine:





After Attaching the volume created from the snapshot, validating the same in the windows server.